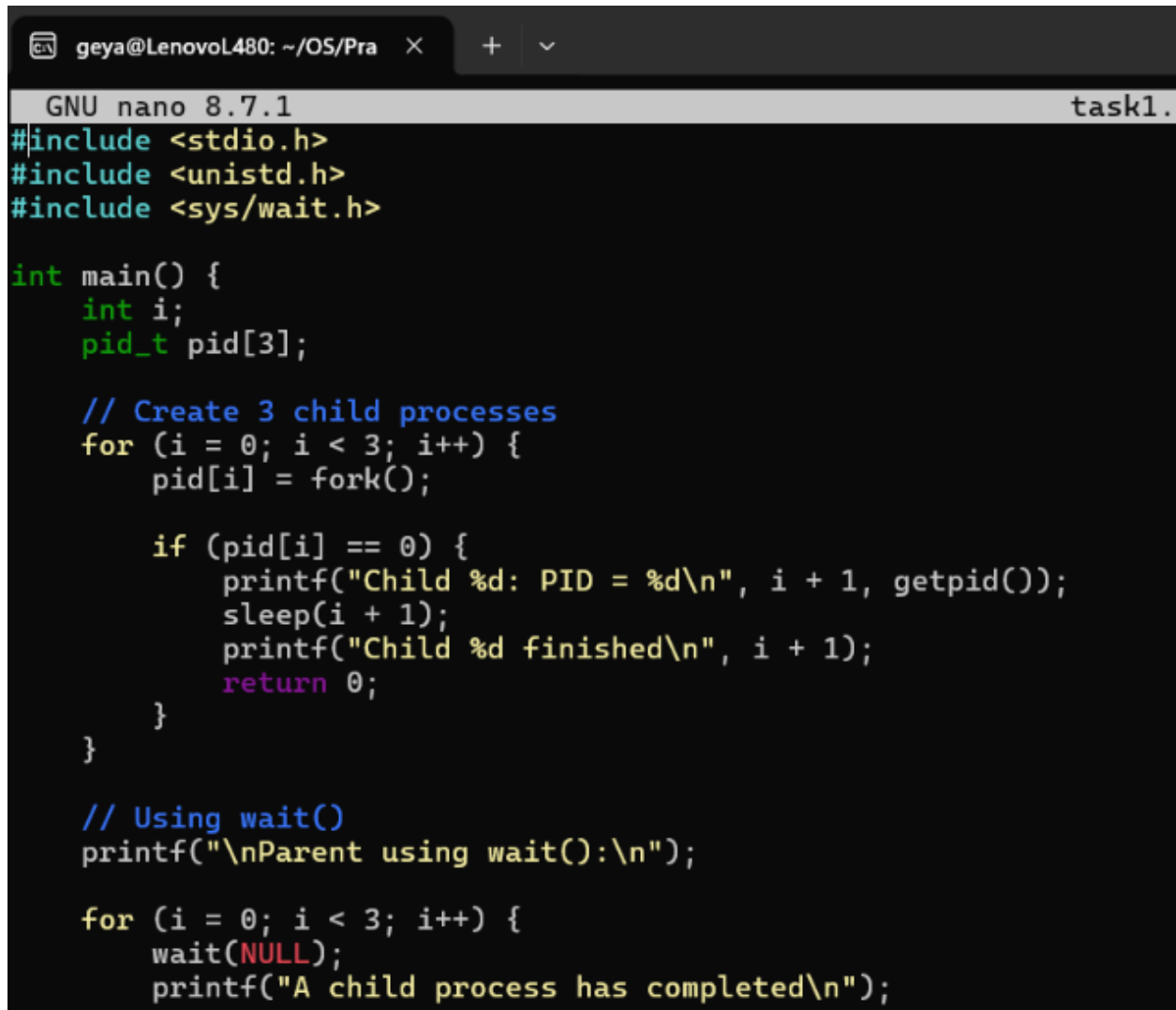


Write a C program where a parent process creates multiple child processes and synchronizes their completion using wait() and waitpid(). Compare the behavior of both functions

Code:



The screenshot shows a terminal window with the nano text editor. The window title is 'geya@LenovoL480: ~/OS/Pra'. The editor is editing a file named 'task1.c'. The code in the editor is as follows:

```
GNU nano 8.7.1 task1.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    int i;
    pid_t pid[3];

    // Create 3 child processes
    for (i = 0; i < 3; i++) {
        pid[i] = fork();

        if (pid[i] == 0) {
            printf("Child %d: PID = %d\n", i + 1, getpid());
            sleep(i + 1);
            printf("Child %d finished\n", i + 1);
            return 0;
        }
    }

    // Using wait()
    printf("\nParent using wait():\n");

    for (i = 0; i < 3; i++) {
        wait(NULL);
        printf("A child process has completed\n");
    }
}
```

```

        printf("A child process has completed\n");
    }

    // Create 3 more children
    for (i = 0; i < 3; i++) {
        pid[i] = fork();

        if (pid[i] == 0) {
            printf("Child %d: PID = %d\n", i + 1, getpid());
            sleep(i + 1);
            printf("Child %d finished\n", i + 1);
            return 0;
        }
    }

    // Using waitpid()
    printf("\nParent using waitpid():\n");

    for (i = 0; i < 3; i++) {
        waitpid(pid[i], NULL, 0);
        printf("Child with PID %d has completed\n", pid[i]);
    }

    return 0;
}
|

```

Output:

```
geya@LenovoL480:~/OS/Practical/Week 4$ ./task1
Child 1: PID = 612
Child 2: PID = 613
```

```
Parent using wait():
Child 3: PID = 614
Child 1 finished
A child process has completed
Child 2 finished
A child process has completed
Child 3 finished
A child process has completed
Child 1: PID = 615
Child 2: PID = 616
```

```
Parent using waitpid():
Child 3: PID = 617
Child 1 finished
Child with PID 615 has completed
Child 2 finished
Child with PID 616 has completed
Child 3 finished
Child with PID 617 has completed
geya@LenovoL480:~/OS/Practical/Week 4$ |
```